

Task3 code:

```
#define LED_PIN 8
#define BUTTON_PIN 2

void setup() {
    pinMode(LED_PIN, OUTPUT);
    pinMode(BUTTON_PIN, INPUT_PULLUP);
    Serial.begin(9600);
}

void loop() {
    int buttonState = digitalRead(BUTTON_PIN);

    if (buttonState == LOW) {
        digitalWrite(LED_PIN, HIGH);
        Serial.println("Button pressed");
    }

    else {
        digitalWrite(LED_PIN, LOW);
        Serial.println("Button released");
    }

    delay(100);
}
```